

Chain-of-Thought (CoT)

Live Exercise on ChatGPT

This worksheet walks you through the same business problem solved three ways: a naive prompt, a basic Chain-of-Thought prompt, and a structured Chain-of-Thought prompt. You will run every prompt on ChatGPT yourself and compare the outputs.

Pick **one** track and stay with it for all three rounds. Then, time permitting, run the other track to see CoT generalize across domains.

How to use this worksheet

1. Open ChatGPT in a new tab.
2. Start a fresh chat for each round so previous reasoning does not leak in.
3. Copy the prompt **exactly** as written.
4. Read the model output. Compare it against the *What to look for* notes.
5. Bring observations back to the room for debrief.

Track A — Finance

Scenario

You are an analyst at a mid-sized retail company. Leadership wants a quick read on whether a proposed price change on a hero SKU will hit the quarterly margin target. You have last quarter's numbers and a few proposed changes. You need a defensible answer.

The shared business context (used in all three rounds)

Context:

- Product: Premium Wireless Earbuds (SKU EB-200)
- Current selling price: Rs 4,000 per unit
- Variable cost per unit: Rs 2,600
- Last quarter units sold: 12,000
- Fixed costs allocated to this SKU per quarter: Rs 48,00,000
- Proposed change next quarter: drop price to Rs 3,600 to compete with a new entrant
- Marketing estimates the price drop will lift units sold by 25%
- Quarterly operating margin target for this SKU: 15%

Round 1 — Naive prompt

Paste this into a fresh ChatGPT chat:

Given the context below, should we make the price change? Answer yes or no and tell me the margin.

Context:

- Product: Premium Wireless Earbuds (SKU EB-200)
- Current selling price: Rs 4,000 per unit
- Variable cost per unit: Rs 2,600
- Last quarter units sold: 12,000
- Fixed costs allocated to this SKU per quarter: Rs 48,00,000
- Proposed change next quarter: drop price to Rs 3,600
- Marketing estimates units sold will lift by 25%
- Quarterly operating margin target: 15%

What to look for in the output:

- A confident yes/no with a single margin number
- No visible working
- You cannot tell where the number came from or whether it is correct
- If two students get different answers, neither can defend theirs

Typical sample output (yours will vary):

Yes, make the price change. With a 25% lift in units, revenue grows and the margin comes in around 17%, above your 15% target. Recommend proceeding.

Round 2 — Basic Chain-of-Thought

New chat. Paste this:

Think step by step before giving your final answer. Show each calculation on its own line.

Question: Given the context below, should we make the price change? Compute the new operating margin and compare it to the 15% target.

Context:

- Product: Premium Wireless Earbuds (SKU EB-200)
- Current selling price: Rs 4,000 per unit
- Variable cost per unit: Rs 2,600
- Last quarter units sold: 12,000
- Fixed costs allocated to this SKU per quarter: Rs 48,00,000
- Proposed change next quarter: drop price to Rs 3,600
- Marketing estimates units sold will lift by 25%
- Quarterly operating margin target: 15%

What to look for in the output:

- Visible arithmetic: new units, new revenue, new variable cost, contribution, operating profit, margin %
- You can now audit each line
- The recommendation is tied to a number you can check

Typical sample output (yours will vary):

Step 1: New units = $12,000 \times 1.25 = 15,000$
Step 2: New revenue = $15,000 \times 3,600 = \text{Rs } 5,40,00,000$
Step 3: New variable cost = $15,000 \times 2,600 = \text{Rs } 3,90,00,000$
Step 4: Contribution = $5,40,00,000 - 3,90,00,000 = \text{Rs } 1,50,00,000$
Step 5: Operating profit = $1,50,00,000 - 48,00,000 = \text{Rs } 1,02,00,000$
Step 6: Operating margin = $1,02,00,000 / 5,40,00,000 = 18.9\%$
Conclusion: 18.9% is above the 15% target. Recommend the price change.

Round 3 — Structured Chain-of-Thought

New chat. Paste this:

Solve this in four labeled steps. Do not skip a step. Do not give the answer until STEP 4.

STEP 1 - RESTATE: List the known numbers and the question in your own words.

STEP 2 - BASELINE: Compute current quarter revenue, variable cost, contribution, operating profit and operating margin.

STEP 3 - SCENARIO: Compute the same five numbers under the proposed price change.

STEP 4 - DECISION: Compare the scenario margin to the 15% target. State the recommendation, two risks of trusting the marketing 25% lift estimate, and one piece of data you would want before committing.

Context:

- Product: Premium Wireless Earbuds (SKU EB-200)
- Current selling price: Rs 4,000 per unit
- Variable cost per unit: Rs 2,600

- Last quarter units sold: 12,000
- Fixed costs allocated to this SKU per quarter: Rs 48,00,000
- Proposed change next quarter: drop price to Rs 3,600
- Marketing estimates units sold will lift by 25%
- Quarterly operating margin target: 15%

What to look for in the output:

- Baseline and scenario are both shown, so you can see *how much* the change moves the needle
- The recommendation now carries caveats: what if the 25% lift does not land?
- You can hand this to a finance lead without re-doing the work

Typical sample output (yours will vary):

STEP 1 - RESTATE: We sell EB-200 at Rs 4,000, variable cost Rs 2,600, last quarter 12,000 units, fixed cost Rs 48,00,000. We want to know if dropping price to Rs 3,600 with an expected 25% unit lift keeps us above a 15% operating margin.

STEP 2 - BASELINE:
Revenue = 12,000 x 4,000 = Rs 4,80,00,000
Variable cost = 12,000 x 2,600 = Rs 3,12,00,000
Contribution = Rs 1,68,00,000
Operating profit = 1,68,00,000 - 48,00,000 = Rs 1,20,00,000
Operating margin = 1,20,00,000 / 4,80,00,000 = 25.0%

STEP 3 - SCENARIO:
Units = 15,000
Revenue = Rs 5,40,00,000
Variable cost = Rs 3,90,00,000
Contribution = Rs 1,50,00,000
Operating profit = Rs 1,02,00,000
Operating margin = 18.9%

STEP 4 - DECISION:
Margin drops from 25.0% to 18.9% but still clears the 15% target, so the change is defensible on paper. Risks: (1) the 25% lift is a marketing estimate, not a tested number - if actual lift is only 10%, margin falls to roughly 13% and misses target; (2) competitor may re-cut price again, eroding the lift. Before committing, I would want a price elasticity read from the last comparable promotion.

Track A debrief — what just happened

Round 1 gave you a number you could not defend. Round 2 gave you a number you could audit. Round 3 gave you a number plus the second-order thinking a finance partner actually needs - baseline vs scenario, named risks, one missing data point. Same model, same context, three very different decision-support artifacts. The difference is the prompt.

Track B — Marketing

Scenario

You run growth at a Direct-to-Consumer skincare brand. A new product launches in three weeks. Last launch underperformed. Leadership wants a launch plan recommendation that is more than vibes.

The shared business context (used in all three rounds)

Context:

- Brand: 'Lumen' - mid-premium DTC skincare, India
- New product: Vitamin C + Niacinamide Serum, Rs 1,200, 30ml
- Target customer: women 25-40, urban, already use serums, follow skincare creators on Instagram
- Channels available: Instagram (organic + paid), email list of 80,000, influencer network of 25 micro-creators (10k-80k followers), brand website with 1.2 lakh monthly visits
- Budget for launch month: Rs 12,00,000
- Last launch result: 1,800 units sold in launch month, against a 4,000 unit target
- Top reason cited for miss: 'category was too crowded, message did not differentiate'
- Goal for this launch: 5,000 units in launch month

Round 1 — Naive prompt

Paste this into a fresh ChatGPT chat:

Given the context below, what should our launch plan be? Give me a recommendation.

Context:

- Brand: 'Lumen' - mid-premium DTC skincare, India
- New product: Vitamin C + Niacinamide Serum, Rs 1,200, 30ml
- Target customer: women 25-40, urban, already use serums
- Channels: Instagram, email list 80,000, 25 micro-influencers, website 1.2L monthly visits
- Budget: Rs 12,00,000
- Last launch: 1,800 units vs 4,000 target. Miss reason: not differentiated
- Goal: 5,000 units in launch month

What to look for in the output:

- A generic 'do Instagram + influencers + email' list
- No diagnosis of *why* last launch failed
- No budget split, no funnel math, no message angle
- Two students will get two different but equally vague plans

Typical sample output (yours will vary):

Launch plan: build hype on Instagram with teaser reels, work with your 25 micro-influencers to create unboxing content, send a launch email to your list with a discount, and run paid ads to retarget website visitors. Make sure the messaging is clear and differentiated. With Rs 12L budget you should be able to hit 5,000 units.

Round 2 — Basic Chain-of-Thought

New chat. Paste this:

Think step by step before recommending. Show your reasoning.

Question: Given the context, propose a launch plan that gets us to 5,000 units. Walk through (a) why the last launch probably missed, (b) what funnel math 5,000 units implies, (c) how the Rs 12,00,000 budget should split across channels, and (d) the single message angle that would differentiate. Then state the recommendation.

Context:

- Brand: 'Lumen' - mid-premium DTC skincare, India
- New product: Vitamin C + Niacinamide Serum, Rs 1,200, 30ml
- Target customer: women 25-40, urban, already use serums
- Channels: Instagram (organic + paid), email 80,000, 25 micro-influencers, website 1.2L monthly visits
- Budget: Rs 12,00,000
- Last launch: 1,800 units vs 4,000 target. Reason: not differentiated
- Goal: 5,000 units in launch month

What to look for in the output:

- A diagnosis of the previous miss, not just a restatement
- Real funnel arithmetic: visits, conversion rate, units
- A budget split with rupee figures, not percentages in the air
- A specific message angle, not 'be differentiated'

Typical sample output (yours will vary):

Step 1 - Diagnosis: 'not differentiated' usually means the product was sold on ingredients (Vit C + Niacinamide) that every competitor also claims. Need an angle that is not about the molecule.

Step 2 - Funnel math: 5,000 units at Rs 1,200 = Rs 60L revenue. At a 2.5% site conversion, 5,000 units needs 2,00,000 product-page visits in the month. Existing 1.2L visits get us 3,000 units at best, so we need ~80,000 incremental qualified visits from paid + influencers + email.

Step 3 - Budget split (Rs 12L total): Instagram paid Rs 5L (retarget + lookalike), Influencer activations Rs 4L (25 creators x ~Rs 16k avg), Email + lifecycle Rs 50k (tooling/creative), Creative production Rs 1.5L, Sampling / first-50-buyers PR kit Rs 1L.

Step 4 - Message angle: instead of 'Vitamin C + Niacinamide', lead with a use-case promise - 'the serum for the 2pm dullness, not the 2am skincare routine'. Position around when it is used, not what is in it.

Recommendation: ship the plan above, hold Rs 1L back for week-2 reallocation to the best-performing channel.

Round 3 — Structured Chain-of-Thought

New chat. Paste this:

Solve this in four labeled steps. Do not skip a step. Do not give the recommendation until STEP 4.

STEP 1 - DIAGNOSE: In your own words, explain why the last launch likely missed, beyond 'not differentiated'. Name two specific failure modes.

STEP 2 - FUNNEL MATH: Starting from the 5,000 unit goal, work backwards to the visits, conversion rate, and incremental traffic the plan needs. State the assumptions you used.

STEP 3 - PLAN: Allocate the Rs 12,00,000 across channels with rupee amounts. For each channel, state what you expect it to contribute in units and why.

STEP 4 - RECOMMEND: Give the final plan. Include the differentiating message angle, two leading indicators you would check in week 1, and the single thing you would cut if the budget got slashed by 30%.

Context:

- Brand: 'Lumen' - mid-premium DTC skincare, India
- New product: Vitamin C + Niacinamide Serum, Rs 1,200, 30ml
- Target customer: women 25-40, urban, already use serums
- Channels: Instagram (organic + paid), email 80,000, 25 micro-influencers, website 1.2L monthly visits
- Budget: Rs 12,00,000
- Last launch: 1,800 units vs 4,000 target. Reason cited: not differentiated
- Goal: 5,000 units in launch month

What to look for in the output:

- A real diagnosis with two named failure modes, not 'crowded category'
- Funnel math you can stress-test
- A budget split tied to expected units per channel, with reasoning
- Leading indicators - the markers of whether the plan is working in week 1, not at the end
- A 'what would I cut' answer, which forces ranked priorities

Typical sample output (yours will vary):

STEP 1 - DIAGNOSE: Two failure modes likely: (1) the launch sold the ingredient stack, which every competitor also sells, so the ad felt like one of many; (2) the existing 1.2L monthly site visitors are mostly returning customers from prior SKUs, so 'launch traffic' was actually the same audience seeing a new product, not new demand entering the funnel.

STEP 2 - FUNNEL MATH: 5,000 units needs ~2,00,000 product-page visits at a 2.5% PDP conversion. Of those, ~50,000 can come from existing site visitors and email (warm), the other ~1,50,000 must come from paid + influencer-driven traffic (cold-to-warm). Assumes Rs 1,200 AOV holds, no bundling, no heavy discount.

STEP 3 - PLAN (Rs 12,00,000): Instagram paid Rs 5,00,000 - retarget + lookalike, target ~1,00,000 incremental visits, expected ~2,500 units - Influencer activations Rs 4,00,000 - 25 micro-creators with use-case content, target ~50,000 qualified visits, expected ~1,200 units - Email + lifecycle Rs 50,000 - 3-email launch flow to 80,000 list, expected ~700 units - Creative + sampling Rs 2,50,000 - PDP video, hero static, first-100-buyer PR kits for earned content, expected ~600 units lift across other channels

STEP 4 - RECOMMEND: Lead with a use-case angle: 'the serum for the 2pm dullness'. Position by occasion, not ingredient. Week-1 leading indicators: (i) PDP conversion rate vs the 2.5% assumption, (ii) cost per qualified site visit from paid Instagram. If budget is cut by 30%, cut the PR kit / sampling spend first - it drives earned lift but is the hardest to attribute, so it dies first in a tight quarter.

Track B debrief — what just happened

Round 1 produced a plan that could fit any brand in any category. Round 2 forced funnel math and a real message angle. Round 3 produced a plan with named risks, leading indicators, and a ranked priority - the thing a growth lead actually argues from. Same prompt skeleton, different domain. The CoT pattern transfers.

Cross-track debrief — the pattern under both tracks

1. Chain-of-Thought is a prompt pattern, not a model setting

Nothing about the model changed between Round 1 and Round 3. You changed the instruction. That is the leverage.

2. Visible reasoning is auditable reasoning

In Round 1 you could not check the answer. In Round 2 you could. That is the difference between a number you defend and a number you hope is right.

3. Structure beats 'think step by step'

'Think step by step' (Round 2) helped. Labeled steps with constraints (Round 3) helped more. The more you specify the shape of the reasoning, the more reliable the output becomes.

4. CoT also surfaces bad inputs

When the model is forced to show working, it sometimes exposes that the question itself is under-specified - a missing number, a vague goal, an assumption that does not hold. A naive prompt hides this. A CoT prompt makes it impossible to hide.

5. Where to use CoT in your real work

Use it whenever the answer must be defended to someone else - finance memos, plan recommendations, diagnostic reads, pricing calls, root-cause analyses. Skip it for one-shot creative tasks where you want the model to be quick and loose.